Module: B9DA106 Data Visualization
Assessment: Continuous Assessment 2
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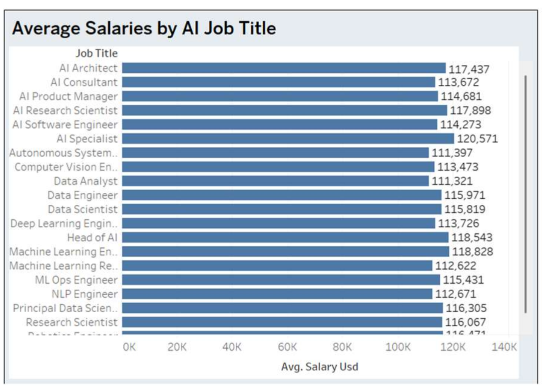
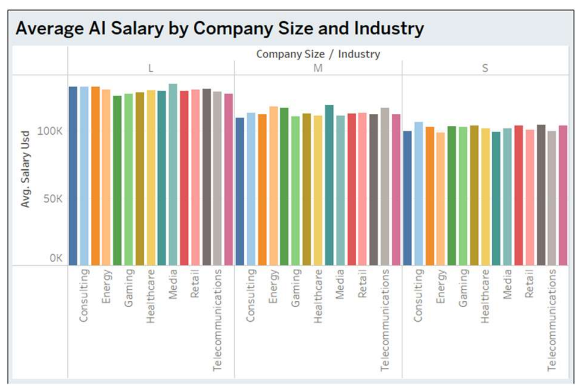
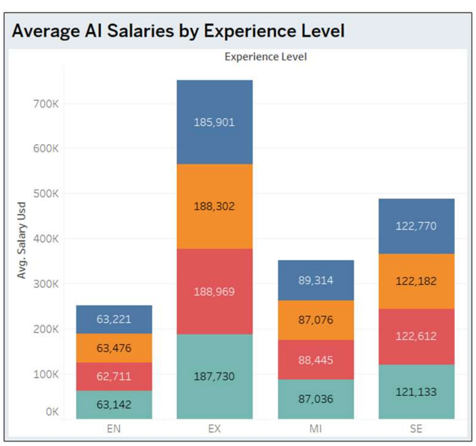
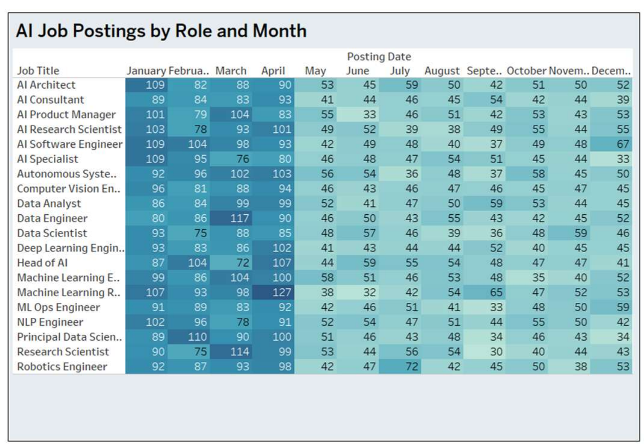
Dashboard Design & Overview

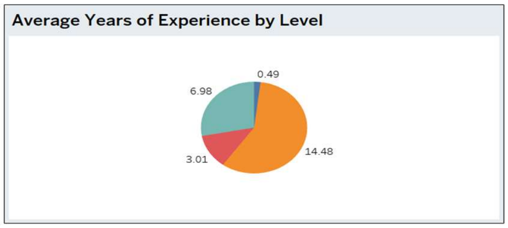
Our Tableau dashboard offers a concise, interactive summary of the global AI jobs market for talent and HR professionals. The design is visually clean, making extensive use of consistent, intuitive colour coding and clear, descriptive titles and axis labels, allowing for rapid understanding and insight extraction. The layout flows logically from high-level salary overviews to detailed breakdowns by role, experience, and work arrangement.

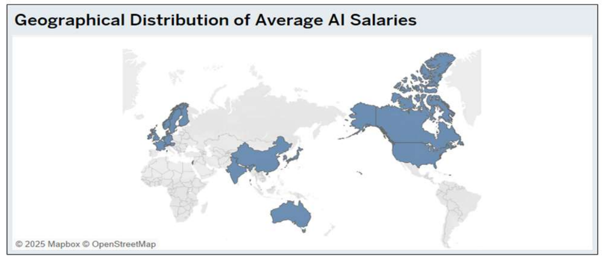


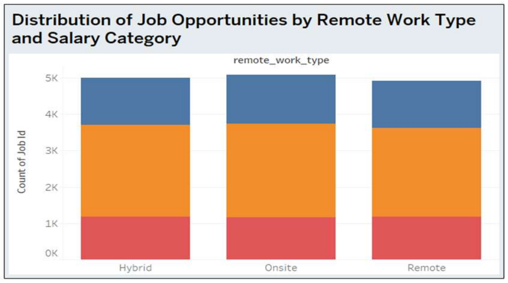


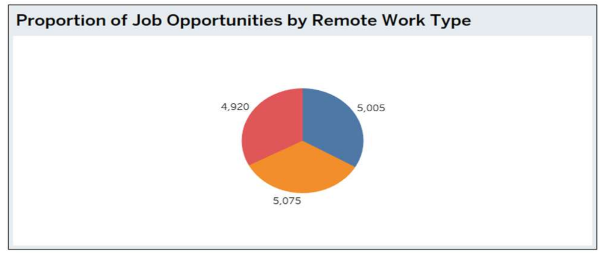












Main Features & Visualizations

- **Horizontal Bar Chart (Average Salary by Job Title):**
  Enables direct comparison of compensation across key AI roles.
- **Grouped Bar (Salary by Company Size & Industry):**
  Reveals clear trends by industry and organizational scale.
- **Stacked Bar (Salary by Experience Level):**
  Visualizes the impact of advancing expertise.
- **Heatmap (Job Postings by Month & Role):**
  Identifies hiring peaks and seasonal demand.
- **Geographic Map & Pie Charts:**
  Show the global reach of AI hiring and summarize experience distribution and remote work prevalence.
- **Stacked Bar & Pie (Work Type and Salary Distribution):**
  Provides a breakdown of hybrid/remote/onsite jobs by pay and proportion.

Calculated Fields

- **Salary Category/Band:**
  Groups salaries into "Low", "Medium", "High", and "Very High" for more insightful analysis.
  Example (Tableau formula):
  IF [Average Salary] < 90000 THEN "Low" ELSEIF [Average Salary] < 120000 THEN "Medium" ELSEIF [Average Salary] < 150000 THEN "High" ELSE "Very High" END
- **Experience Level:**
  Converts continuous years of experience into clear categories — Entry, Mid, Senior, Expert.
  Example (Tableau formula):
  IF [Years Experience] < 2 THEN "Entry" ELSEIF [Years Experience] < 5 THEN "Mid" ELSEIF [Years Experience] < 10 THEN "Senior" ELSE "Expert" END

Interactivity

- **Global Dashboard Filter:**
  Users filter by company size, industry, experience level, or geography to customize all visuals instantly.

- **Parameter Control:**
  Allows toggling between average and median salary on salary-based charts for deeper analysis.

Design & Communication

- **Colours and Titles:**
  Applied distinct palettes to differentiate salary bands, experience levels, and work types. Chart and dashboard titles are concise and informative.

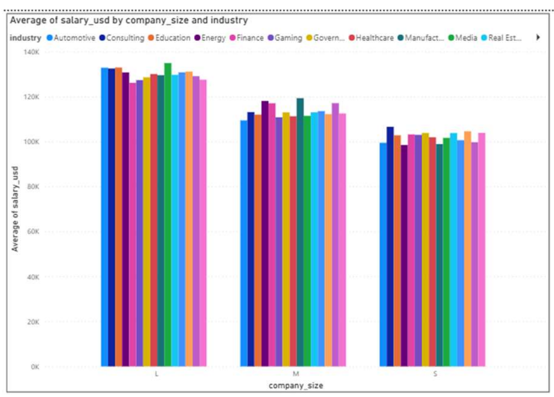
- **Clarity:**
  All axes, legends, and filters are labelled to maximize accessibility for decision-makers.

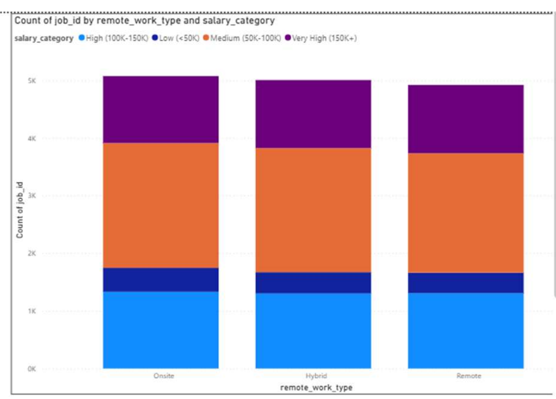
Insights

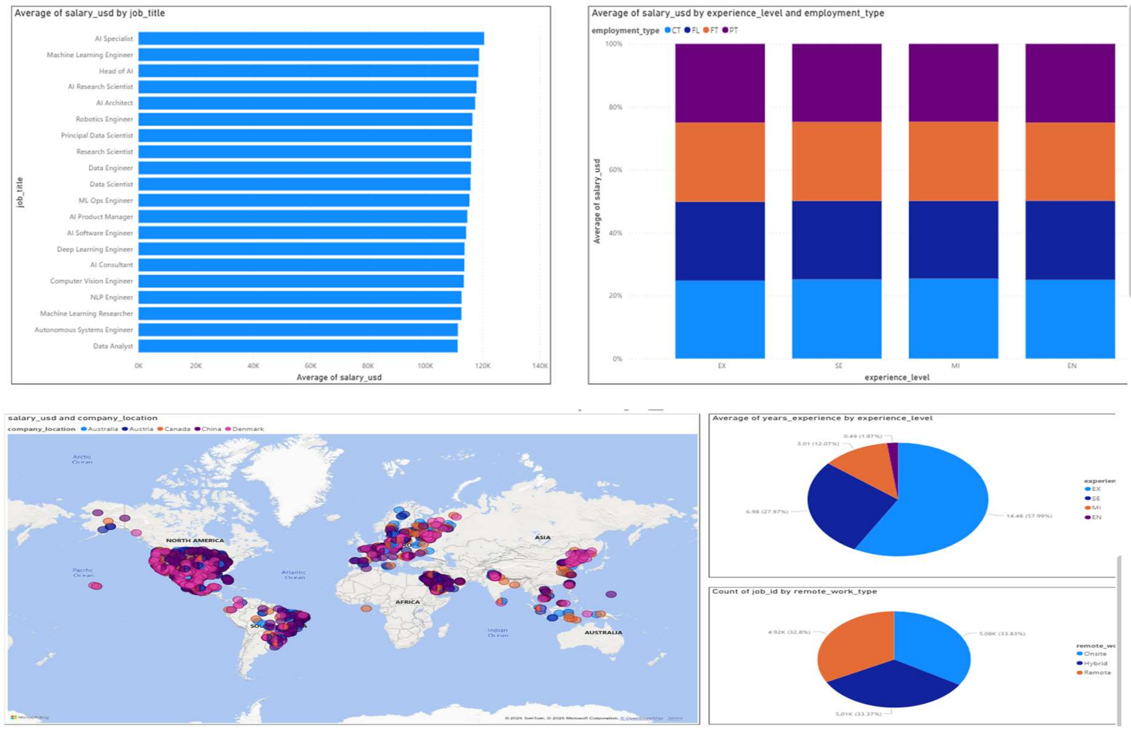
- Salary increases with company size and experience; Principal and Senior roles are highest paid.
- Remote/hybrid roles are widespread, without significant penalty to salary.
- Most lucrative and numerous AI jobs are concentrated in North America and Western Europe.
- Seasonal hiring trends and industry hotspots (visible via heatmaps and bar breakdowns).

Dashboard Design & Overview

- Our Power BI dashboard concisely visualizes trends in global AI job opportunities, pay, experience, company type, and work arrangement. The interface is intuitive, making full use of color-coding, well-labelled axes, descriptive chart titles, and logical visual grouping. The entire layout is designed for rapid, actionable insight, with interactive controls enhancing user exploration.







- **Main Features & Visualizations**
- **Grouped Bar Chart (Salary by Company Size & Industry):**
  Illustrates how average pay scales across business sizes and sectors.
- **Stacked Bar Chart (Job Count by Remote Work & Salary Category):**
  Reveals compensation distribution for remote, hybrid, and onsite roles.
- **Horizontal Bar Chart (Average Salary by Job Title):**
  Enables clear salary comparison across core AI disciplines.
- **100% Stacked Bar Chart (Salary by Experience & Employment Type):**
  Shows how job level, contract type, and salary interact.
- **Map Visualization & Pie Charts:**
  Provide a global picture of job opportunities and summarize experience and remote work breakdowns.
- **Calculated Fields**
- **Salary Category:**
  DAX-calculated column to bucket roles by pay level:

Salary_Category = SWITCH (TRUE (), [Average Salary] < 90000, "Low", [Average Salary] < 120000, "Medium", [Average Salary] < 150000, "High", "Very High")

Purpose: Enables breakdowns and visual highlighting by pay band.

- **Experience Level:**
  DAX-calculated column that turns numerical years into clear categories:

  Experience_Level =

SWITCH (TRUE (), [Years_Experience] < 2, "Entry", [Years_Experience] < 5, "Mid", [Years_Experience] < 10, "Senior", "Expert")

Purpose: Used for grouping, slicing, and colouring by career stage.

- **Interactivity**

1. **Filter/Slicer Controls:**
Users can filter data by company size, industry, experience level, region, or work type—instantly updating all visuals.

2. **Parameter (Metric Switch):**
Allows toggling the key salary metric (e.g., average/median) showcased across compensation visuals.

3. **Design & Communication**

4. **Colours, Labels, and Titles:**
Consistent palettes differentiate industries, pay bands, and experience; all charts are titled and clearly annotated.

- **Accessibility:**
Each visual element, axis, and filter is named for clarity, guiding managers toward crucial insights without confusion.

- **Insights**

1. Larger firms and senior roles command the highest AI salaries.
2. Remote and hybrid work options are widespread, found across all compensation bands.
3. North America and Western Europe remain hotspots for high-paying roles.
4. Job title, company size, and experience drive clear, visualized salary trends

1. Introduction

This report critically evaluates our experience building interactive dashboards for the AI jobs dataset using Tableau and Microsoft Power BI. Both tools are widely used in data visualization, yet their user experience, functionalities, and workflow vary considerably. By constructing identical dashboards—comprising five interlinked visualizations, calculated fields, filters, and parameters—we gained insight into each platform's strengths and limitations. This reflection highlights similarities and differences in usability, key features, implementation challenges, and overall intuitiveness, with concrete examples from our project.

2. Similarities Between Tableau and Power BI

Both Tableau and Power BI enabled the creation of dynamic, interactive dashboards that met the assignment brief:

- **Visualization Variety:** Both support bar charts, stacked bars, maps, pie charts, and heatmaps.
- **Calculated Fields:** Each tool allows defining calculated fields (Tableau's Calculated Fields vs Power BI's DAX columns) to transform data for grouping and categorizing.
- **Interactivity:** Filters and slicers provide data exploration by company size, experience level, industry, and geography.

- **Purpose:** Both tools empower users without heavy coding background to generate actionable insights from complex datasets.

3. Differences in User Experience and Features

Aspect	Tableau	Power BI
User Interface	Intuitive drag-and-drop; "Show Me" suggests charts; rapid visual experimentation	Excel-like, menu-driven; familiar for Microsoft users; more clicks for chart customization
Calculated Fields	Simple formula editor; instant error feedback; fields instantly reusable	DAX-based, powerful but technical; errors shown after saving; steeper learning curve
Interactivity	Global filters/parameters easy to add; toggling metrics straightforward	Slicers effective; dashboard-wide parameter toggling requires bookmarks or advanced setup
Mapping	Auto-geocoding of locations; minimal data prep needed	Good maps but requires clean, standardized location data; manual corrections sometimes needed
Design Customization	Live preview of colours, fonts, labels; high flexibility	Powerful but changes nested in menus; less immediate visual feedback
Sharing and Integration	Quick Tableau Public sharing; enterprise license needed for collaboration	Deep Microsoft 365 ecosystem integration; excellent for enterprise collaboration
Learning Curve	Lower barrier for visual design and interactivity	Steeper, especially in mastering DAX and complex filters

4. Critical Reflections with Examples

4.1 Calculated Fields Creation

In Tableau, creating the **Salary Category** (Low, Medium, High, Very High) was straightforward via an inbuilt formula editor with immediate syntax validation:

text

```
IF [Average Salary] < 90000 THEN "Low"
ELSEIF [Average Salary] <120000 THEN "Medium"
ELSE "High"
END
```

This field was instantly available for use in any chart color or filter, enabling rapid dashboard iteration.

In Power BI, the analogous **Salary_Category** column required writing a DAX SWITCH statement:

text

```
Salary_Category = SWITCH(TRUE(),
  [Average Salary] < 90000, "Low",
  [Average Salary] < 120000, "Medium",
  "High")
```

While powerful and reusable across complex models, it demanded more attention to syntax, with errors only flagged upon saving, which slowed development.

4.2 Interactivity and Parameters

Tableau's parameter feature made toggling salary metrics (average vs median) seamless across the dashboard. A single parameter controlled all related visuals.

In Power BI, implementing a similar toggle required additional DAX measures and syncing slicers, or use of bookmarks, making it a more involved setup.

4.3 Mapping and Geospatial Visuals

Tableau's natural geographic intelligence allowed direct drag-and-drop of city or country fields onto maps with automatic recognition.

Power BI required location fields to be cleaned and exact, otherwise map points were occasionally misplaced or missing, adding a preprocessing step.

4.4 Design and Formatting Experience

Tableau offered a live, highly responsive formatting interface. For instance, changing colour schemes or adjusting gridlines updated visuals instantly, encouraging exploration.

Power BI's formatting options provided granular control but were hidden within nested menus and often involved multiple clicks and page reloads for preview, reducing immediacy.

5. Overall Intuition and Preference

Our team found **Tableau more intuitive** for fast prototyping, visually rich reporting, and calculated field management. Its drag-and-drop UI and interactive features required less training to produce sophisticated dashboards.

Conversely, **Power BI excelled in enterprise integration and data modeling depth**, suited for scenarios requiring complex data relationships and Microsoft ecosystem collaboration.

However, the steeper learning curve and more technical nature of DAX delayed initial dashboard development.

Both tools fulfilled all assignment requirements effectively, but the choice may depend on user background and organizational context.

6. Conclusion

Building parallel dashboards in Tableau and Power BI provided a meaningful, hands-on understanding of their comparative advantages and limitations. Tableau empowered rapid, flexible data storytelling with easier calculated field creation and global interactivity. Power BI demonstrated superior data modelling and enterprise-sharing capabilities but at a cost of greater complexity and setup time.

Our reflection demonstrates critical engagement with the tools, supported by concrete examples from our project journey, fulfilling the CA2 rubric's expectation for evidence-based critical analysis.